HackerRank Problem Notes

Q7) Min Max sum
Total sum - arr[min] and Total sum - arr[max]

Q9) Time Conversion
Extract all substring in respective and if pm, hour+12
Edge case – 12 am and 12 pm
Integer.parseInt() String.valueOf()

Q10) grading student
Create a new function to handle all the cases

Q11) Orange n Apple
maths

Q12) Number line jump
X1 == x2 && if (v1==v2)=> x1==x2
V1-v2 % x2-x1 == 0

Q13) Breaking record
A[i] = max,min

Q14) subarray division
Sliding window

Q15) Divisible sum by P
Easy

Q16) Migratory BIrd
Hw – try hashmap

Q18) Drawing book
Check gpt and hackerrank

Q19) Drawing Book
First = n/2
End = n/2 - p/2
Return Math.min(first,end)

Q21) Cat n Mouse
Easy

Q22) Day of the programmer

Georgian → x%4 && x%100!=0 || x%400==0

Julian → x%4==0

if(x==1918) return “"26.09.1918"

Q23) Counting Valleys

U → count++   {count==0 → valley++}

L → count–

Q24) Hurdle race

Find array max

Return (max>k) max-k : 0

Q25) Designer pdf

Find max value from array using letters

Height = word.charAt(i)-’a’ // index of every char

Max = max(max, h.get(height) // max height from array

Max * word.length();

Q26) Electronics Shop

Easy

Q27) Utopian Tree

h%2!=0 height*2=0

Else height++

Q28) Angry Professor

Easy

Q29) Beautiful Days at the Movie

Easy (st - calculate reverse)%k

Q30) Viral Advertising

Little math logic

Q31) Cut the Stick

Collection logic + arraylist

Q33) Find digit

Easy

Q32) Sequence Equation

HashMap

Q34) Circular Array Rotation

(q-(k%n)+n) % n

Q35) Missing Number

Frequency hashmap → collections.sort(list)

Q36) Find the median

n%2==0 → arr.get(n/2)

n%2!=0 → arr.get ((n/2 + n/2+1)/2)

Q37) Closet Number

Sort arr -> Math.abs(arr.get(i) - arr.get(i-1)

If (this < max) { list.clear() max = this list.add(i-1,i)

Else if (this == max) { list.add(i-1,i)};

Q38) Sherlock and Array

Calculate total sum

for(int i : arr) if(leftsum == totalsum-totalsum-i) “Yes” ; leftsum+=i

Return “No”;

Q39) Jim and the orders

TreeMap does not allow duplicate values

Use List<int[]> → Collections.sort(list, (a,b) → a[0] - b[0])

for(int[] i : list) al.add(list.get(i[1]);

Q40) Mark and Toys

```
Sort arr;
Int i=0, sum=0;
for( i=0; i<arr.size(); i++){
        if(sum + prices.get(i)>k) break;
        sum+= prices.get(i);
}
return i;   // why not i+1 ? as for loop while stop at i+1 so its auto updated
```

Q41) Challenge 1 (binary search)

Write binary search

Q42) Challenge 2 (insertion sort p1)

```
int key = arr.get(arr.size()-1);
int i = arr.size()-2;
while(i>=0 && arr.get(i)>key){
    arr.set(i+1,arr.get(i));
    printArray(arr);
    i--;
}
arr.set(i+1,key);
printArray(arr);
```

Q43) Challenge 3 (insertion sort p2)

For every second loop complete → print all element of arr

```
for(int k=0;k<n;k++){
  System.out.print(arr.get(k)+" ");
}
System.out.println();
```

Q44) Challenge 4 (Run time of an algorithm)

In second loop whenever there's swap count++;

```
public static void swap(List<Integer> arr, int i, int j){
    int temp = arr.get(i);
    arr.set(i, arr.get(j));
    arr.set(j, temp);
}
```

Q45) Halloween Sale

Math logic + while loop

```
Int count=0;
while(s>=p)
        s-=p;
        count++;
        p = Math.max(p-d, m);
return count;
```